

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 4

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 5 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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PHYS10302 Vibrations and Waves

Week 4 Questions

These problems offer practice in solving differential equations relevant to vibrations.

1. Solve the quadratic equation

$$r^2 + 4r + 4 = 0.$$

2. Suppose a mass of 1 kg stretches a spring of spring constant 192 N/m in the absence of damping. If the mass is displaced down a further 2 m and set in motion with an upward velocity of 1 m/s, write down and solve the differential equation for the motion of the mass and determine the position of the mass at time t in terms of $\sin$ and $\cos$ solutions. Choose positive y as upwards from equilibrium. What is the period of the motion?

3. Suppose a mass on a spring satisfies the differential equation

$$\frac{d^2y}{dt^2} + \frac{dy}{dt} + 1.25y = 2 \cos(t),$$

with initial conditions

$$y(0) = 2, \quad \frac{dy}{dt}(0) = 3.$$

Find the solution to this initial value problem by finding the complementary function and the particular integral. For the complementary function, solve the homogeneous equation and write the solutions as exponentials. Then, take suitable linear combinations of these solutions to express them in terms of $\sin$ and $\cos$ functions. What happens to the solution of the initial value problem at large t ?

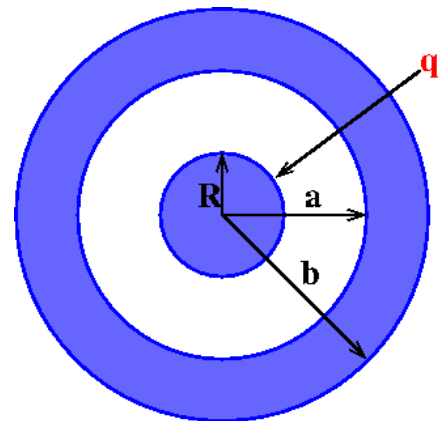
E&M Tutorial Sheet 4: Conductors

- 1) (Revision) The scalar electric potential in a region of space is found to be $\phi(r) = K(a^2 - r^2)$ for $r < a$ and $\phi(r) = 0$ for $r \geq a$. Calculate the electric field in the regions $r < a$ and $r > a$.

Here are some questions about fields, charges and conductors. Read back through the properties of conductors that you were given in lectures. These questions require careful application of those results to some more complicated situations.

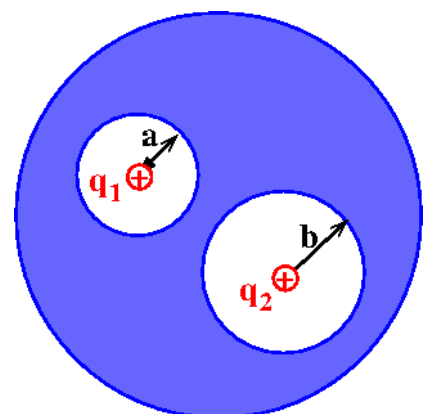
- 2) Consider a volume of conductor. What is the electric potential difference between two points within the conductor? What is the electric field strength within the conductor? What is the density of charge within the volume of the conductor? If the conductor is charged, where does the charge reside? Justify your answers.
- 3) A conducting sphere of radius R carries a total charge of q . It is surrounded by an electrically neutral conducting shell with an inner radius, a , and outer radius, b .

- (a) What is the surface charge density on the sphere, the inner surface of the shell and the outer surface of the shell?
- (b) What is the electric field in the following regions:
(i) $r < R$, (ii) $R < r < a$, (iii) $a < r < b$, (iv) $r > b$?
- (c) Find the electric potential at the centre of the sphere taking infinity as the zero reference point.
- (d) Sketch E and ϕ as a function of r .
- (e) If the outer surface of the shell is grounded so that it is held at a potential of zero, how do the answers to (a) and (b) change?



- 4) A solid, electrically neutral metallic sphere radius R has two spherical cavities of radii a and b . Point charges, $+q_1$ and $+q_2$, are placed at the centre of each cavity as shown in the diagram.

- a) Find the surface charge densities on the inner surface of each cavity and on the outer surface of the sphere.
- b) What is the electric field outside the sphere?
- c) What is the electric field in each cavity?
- d) What is the force on the charges q_1 and q_2 ?
- e) Which of the answers a) to d) change if a third charge, q_3 , is brought up close to the sphere?
- f) Describe qualitatively what will happen if one of the charges is placed slightly off-centre inside its cavity.



- 5) Two solid metallic spheres with radii R_1 and R_2 are fixed a large distance apart (large enough that electric fields due to one have a negligible influence on the other). Sphere 1 is charged with a (positive) charge Q while sphere 2 is neutral.
- a) Calculate the electric potential, ϕ , of sphere 1 relative to a point at infinity.

- b) The two spheres are now connected together by a thin wire. Describe what happens.
Hint: What happens when a wire connects two points at different potentials?
- c) Once the system has reached equilibrium, the wire is removed. Ignoring any small amount of charge that remains on the wire, calculate:
- (i) the charges, Q_1' and Q_2' ;
 - (ii) charge densities, σ_1' and σ_2' ;
 - (iii) potentials, ϕ_1' and ϕ_2' in terms of ϕ ;
 - (iv) electric field strengths at the surfaces, E_1' and E_2' , of the two spheres.
- d) Calculate the total potential energy of the system before and after connecting the spheres. Discuss the energy changes that occur.

PHYS10372 Mathematics 2: Tutorial-04 Questions

These questions cover the material from week-4 of PHYS10372: calculating fluxes using the div operator and the divergence theorem.

1. Choose an appropriate coordinate system and calculate the net flux of the following vector fields across the following surfaces. Note you should find the flux using surface integrals (not using the divergence theorem volume integral).

(a) the vector field: $\mathbf{v} = x^2 \hat{\mathbf{k}}$ out of the region enclosed by the hemisphere $x^2 + y^2 + z^2 = R^2 (z > 0)$ and the xy plane.

(b) the vector field: $\mathbf{v} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}$ out of a cube of sides L with vertices at $(0,0,0)$, $(0, L, 0)$, $(0, L, L)$, (L,L,L)

(c) the vector field: $\mathbf{v} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}$ out of a cylinder of radius and height R whose axis lies along the z -axis and whose base is on the xy plane.

(d) the vector field: $\mathbf{v} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}$ out of a sphere centred at the origin.

[Answer: (a) 0; (b) $3L^3$; (c) $3\pi R^3$; (d) $4\pi R^3$]

You may need to evaluate terms such as: $\hat{\mathbf{r}} \cdot \hat{\mathbf{i}}$ in spherical polars as follows:

$$\hat{\mathbf{r}} \cdot \hat{\mathbf{i}} = \frac{\mathbf{r} \cdot \hat{\mathbf{i}}}{r} = \frac{x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}}{r} \cdot \hat{\mathbf{i}} = \frac{x}{r} = \frac{r \sin \theta \cos \phi}{r} = \sin \theta \cos \phi$$

2. Calculate the flux of the vector field $\mathbf{F} = 3x \hat{\mathbf{i}}$ out of a sphere of radius a centred at the origin, first as a flux (area) integral, $\iint \mathbf{F} \cdot d\mathbf{A}$ and then check your answer via a volume integral using the divergence theorem i.e. $\iiint \nabla \cdot \mathbf{F} dV$

It is easiest to tackle this problem using spherical polar co-ordinates – draw a diagram and make sure you have the correct volume and surface element terms in your integrals.

[Answer: $4\pi a^3$]